

A Direct Ackermann Comparison for the Seidel–Sharir Top-Down Path-Compression Hierarchy

Path-compression proof-digestion note

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Abstract

This note gives a source-faithful, proof-digested presentation of how the Seidel–Sharir top-down path-compression hierarchy yields an inverse-Ackermann bound under an explicit Ackermann normalization. The central comparison is phrased through the integer-threshold inverse

$$R_k(t) = \max\{r \geq 0 : J_k(r) \leq t\} \quad (t \in \mathbb{N}),$$

and proves

$$R_{z+1}(Q) \geq A(z, 4Q)$$

for all integers $z \geq 1$ and $Q \geq 1$. Thus the comparison target $R_{cz+C}(Q) \geq A(z, DQ)$ holds with constants $c = 1$, $C = 1$, and $D = 4$. The proof stays within the Seidel–Sharir top-down J -hierarchy and the associated $g^\diamond$ recurrence; it does not use a bottom-up Tarjan-potential argument. Translating the comparison to the packet normalizations $L(n) = \lceil \log_2 \max(n, 2) \rceil$ and $Q(m, n) = \lceil 1 + m/n \rceil$ gives $\alpha_J^Q(m, n) \leq \alpha_Q(m, n) + 1$, the source-faithful real-threshold bound $\alpha_J^S(m, n) \leq \alpha_Q(m, n) + 2$ with a +1 improvement in the integral case, and the cost bound

$$f(m, n, L(n)) \leq (\alpha_Q(m, n) + 3)m + 4n.$$

1 Introduction

Seidel and Sharir gave a top-down analysis of path compression in union-find structures, deriving recurrences from which the usual bounds arise naturally; their abstract emphasizes that the inverse-Ackermann bound can be derived without introducing the Ackermann function itself [?]. The Dagstuhl Seminar 25191 report records Tarjan’s later comparison question about the top-down analysis in Section 5.3 of the seminar report [?]. Classical union-find analysis and the Ackermann hierarchy enter through Tarjan’s work on set union [?].

This note addresses that comparison problem at the level of proof exposition and digestion. It gives a direct comparison from the Seidel–Sharir top-down J -hierarchy to a concrete Ackermann normalization. The point is not to claim a new union-find upper bound. Rather, the contribution is a source-anchored proof that the Seidel–Sharir J hierarchy itself implies an inverse-Ackermann bound, with all constants and normalizations explicit.

The main theorem is the threshold comparison

$$R_{z+1}(Q) \geq A(z, 4Q) \quad (z \geq 1, Q \geq 1),$$

where, for integer thresholds $t \geq 0$,

$$R_k(t) = \max\{r \geq 0 : J_k(r) \leq t\}.$$

Thus the target comparison $R_{cz+C}(Q) \geq A(z, DQ)$ holds with

$$c = 1, \quad C = 1, \quad D = 4.$$

The rest of the paper translates this comparison into the packet-normalized alpha bounds and the resulting cost estimate.

What is new here? This note does not claim a new union-find bound. It gives a direct, source-anchored proof that the Seidel–Sharir top-down J hierarchy implies an inverse-Ackermann bound under an explicit Ackermann normalization, addressing Tarjan’s Dagstuhl comparison problem at the level of proof exposition and digestion.

2 Source setup and conventions

This section records the source facts and packet normalizations used in the proof. The source facts are the top-down path-compression recurrence machinery of Seidel and Sharir [?]; the later proof only compares the resulting J hierarchy with an Ackermann hierarchy.

2.1 The transforms g^* and $g^\diamond$

For an integer-valued function g satisfying $g(r) < r$ for $r > 0$, Seidel and Sharir define an iteration count g^* by

$$g^*(r) = \begin{cases} 0, & r \leq 1, \\ 1 + g^*(g(r)), & r > 1, \end{cases}$$

and use the related transform

$$g^\diamond(r) = \begin{cases} g(r), & g(r) \leq 1, \\ 1 + g^\diamond(\lceil \log_2 g(r) \rceil), & g(r) > 1. \end{cases}$$

All logarithms in this paper are base 2.

2.2 The J hierarchy

Seidel–Sharir Corollary 6 defines the hierarchy

$$J_0(r) = \left\lceil \frac{r-1}{2} \right\rceil, \quad J_{k+1} = J_k^\diamond.$$

Equivalently,

$$J_{k+1}(r) = \begin{cases} J_k(r), & J_k(r) \leq 1, \\ 1 + J_{k+1}(\lceil \log_2 J_k(r) \rceil), & J_k(r) > 1. \end{cases}$$

2.3 The source-correct shifting lemma and recurrence

The source-correct top-down shifting lemma is Seidel–Sharir Lemma 5:

If

$$f(m, n, r) \leq km + 2ng(r),$$

then

$$f(m, n, r) \leq (k + 1)m + 2ng^\diamond(r).$$

Iterating this lemma through the J hierarchy gives the source recurrence consequence

$$f(m, n, r) \leq km + 2nJ_k(r).$$

The factor $2n$ is part of the source statement and is used throughout the cost translation.

2.4 Rank and threshold normalizations

The source rank cutoff is governed by the rank bound $\log_2 n$. The packet convention stabilizes the cutoff as

$$L(n) = \lceil \log_2 \max(n, 2) \rceil.$$

For $n \geq 2$, this agrees with $\lceil \log_2 n \rceil$; for $n = 1$, it avoids a degenerate zero cutoff.

The source inverse threshold is

$$J_k(\lceil \log_2 n \rceil) \leq 1 + \frac{m}{n}.$$

The packet integer threshold is

$$Q(m, n) = \left\lceil 1 + \frac{m}{n} \right\rceil.$$

This is a normalization device: it turns the real threshold $1 + m/n$ into an integer threshold suitable for the threshold inverse R_k .

Remark 2.1 (The non-source-faithful $2m/n$ threshold). *The threshold $J_k(\lg n) \leq 2m/n$ is not the source threshold recorded in the current packet. It is treated below only as a quarantined alternate target; the source-faithful threshold is $1 + m/n$.*

3 Blackboard proof idea

The proof has one moving part: the $\diamond$ recurrence turns a bound at level k into a much larger threshold inverse at level $k + 1$.

For integer thresholds $t \geq 0$, define

$$R_k(t) = \max\{r \geq 0 : J_k(r) \leq t\}.$$

If $r \leq R_k(2^B)$, then $J_k(r) \leq 2^B$. In the recursive branch of $J_{k+1} = J_k^\diamond$,

$$J_{k+1}(r) = 1 + J_{k+1}(\lceil \log_2 J_k(r) \rceil),$$

and the logarithmic argument is at most B . Therefore, if $B \leq R_{k+1}(t)$, then the recursive call costs at most t , and

$$J_{k+1}(r) \leq t + 1.$$

This gives the threshold recurrence

$$R_{k+1}(t + 1) \geq R_k(2^{R_{k+1}(t)}).$$

That recurrence has Ackermann flavor: increasing the J -level by one lets us feed an exponential of a previous threshold into the lower-level inverse.

The proof turns this into the invariant

$$R_j(x) \geq A(j + 1, x) \quad (j \geq 1, x \geq 1).$$

The base level $R_0(t) = 2t + 1$ is too small to dominate $A(1, t) = 2^t$; this is why the comparison has a one-level shift. After the invariant is proved, a threshold jump gives

$$R_{z+1}(Q) \geq A(z, 4Q).$$

4 Technical proof

4.1 Ackermann normalization and threshold inverse

Definition 4.1 (Ackermann normalization). *Define $A(i, x)$ by*

$$A(0, x) = 2x,$$

and, for $i \geq 1$,

$$A(i, 0) = 0, \quad A(i, 1) = 2, \quad A(i, x) = A(i - 1, A(i, x - 1)) \quad (x \geq 2).$$

In particular, $A(1, x) = 2^x$ for $x \geq 1$.

Definition 4.2 (Threshold inverse). *For an integer threshold $t \geq 0$, define*

$$R_k(t) = \max\{r \geq 0 : J_k(r) \leq t\}.$$

The maximum is finite by Lemma 4.4 below.

4.2 The basic J_k package

Lemma 4.3 (Diamond preservation). *Let $g : \mathbb{N} \rightarrow \mathbb{N}$ be integer-valued, nonnegative, nondecreasing, unbounded, and satisfy*

$$g(0) = 0, \quad g(r) < r \quad (r > 0).$$

Define $h = g^\diamond$ by

$$h(r) = \begin{cases} g(r), & g(r) \leq 1, \\ 1 + h(\lceil \log_2 g(r) \rceil), & g(r) > 1. \end{cases}$$

Then h is integer-valued, nonnegative, nondecreasing, unbounded, satisfies $h(0) = 0$, and satisfies

$$h(r) \leq g(r) \quad (r \geq 0),$$

hence

$$h(r) < r \quad (r > 0).$$

Proof. For every integer $x \geq 2$,

$$\lceil \log_2 x \rceil \leq x - 1.$$

Thus, in the recursive branch $g(r) > 1$, with $u = \lceil \log_2 g(r) \rceil$, we have

$$u \leq g(r) - 1 < r.$$

So $h(r)$ is defined by recursion on smaller arguments. It is integer-valued and nonnegative because g is. Also $g(1) = 0$, since $g(1) < 1$ and g is nonnegative.

We prove $h(r) \leq g(r)$ by induction on r . If $g(r) \leq 1$, then $h(r) = g(r)$. If $g(r) > 1$, set $u = \lceil \log_2 g(r) \rceil$. Since $u < r$, induction gives $h(u) \leq g(u)$. Since $g(0) = 0$ and $g(u) < u$ for $u > 0$, we have $g(u) \leq u$ for every u . Thus

$$h(r) = 1 + h(u) \leq 1 + u \leq g(r).$$

This proves $h(r) \leq g(r)$, and hence $h(r) < r$ for $r > 0$.

It remains to prove monotonicity. We prove by strong induction on s that $r \leq s$ implies $h(r) \leq h(s)$. Let $r \leq s$, and set $x = g(r)$ and $y = g(s)$. Since g is nondecreasing, $x \leq y$. If $y \leq 1$, then $h(r) = x \leq y = h(s)$. If $y > 1$ but $x \leq 1$, then $h(r) = x \leq 1 \leq h(s)$. If $x > 1$, put

$$u = \lceil \log_2 x \rceil, \quad v = \lceil \log_2 y \rceil.$$

Then $u \leq v$ and $v \leq y - 1 < s$. By strong induction, $h(u) \leq h(v)$. Therefore

$$h(r) = 1 + h(u) \leq 1 + h(v) = h(s).$$

Thus h is nondecreasing.

Finally, h is unbounded. We prove by induction on $M \geq 0$ that there exists r with $h(r) \geq M$. For $M = 0$ this is immediate. Suppose it holds for M , and choose u with $h(u) \geq M$. Since g is unbounded, choose r with $g(r) > 2^u$. Then $g(r) > 1$ and $v = \lceil \log_2 g(r) \rceil > u$. By monotonicity, $h(v) \geq h(u) \geq M$. Hence

$$h(r) = 1 + h(v) \geq M + 1.$$

This proves unboundedness. □

Lemma 4.4 (Basic J_k package). *For every $k \geq 0$:*

- (i) J_k is integer-valued and nonnegative on $\mathbb{N}$;
- (ii) $J_k(0) = 0$;
- (iii) J_k is nondecreasing;
- (iv) J_k is unbounded on $\mathbb{N}$;
- (v) $J_k(r) < r$ for every $r > 0$, in particular for every $r > 1$;
- (vi) $J_{k+1}(r) \leq J_k(r)$ for every $r \geq 0$;
- (vii) $R_k(t)$ is finite for every integer $t \geq 0$;

(viii) for integer thresholds $0 \leq t \leq t'$, we have $R_k(t) \leq R_k(t')$, and $R_{k+1}(t) \geq R_k(t)$.

Proof. For $k = 0$,

$$J_0(r) = \left\lceil \frac{r-1}{2} \right\rceil.$$

This is integer-valued and nonnegative for integer $r \geq 0$. It satisfies $J_0(0) = 0$, is nondecreasing, is unbounded, and satisfies $J_0(r) < r$ for $r > 0$.

Assume J_k has the first five properties. Apply Lemma 4.3 to $g = J_k$. Since $J_{k+1} = J_k^\diamond$, it follows that J_{k+1} is integer-valued, nonnegative, nondecreasing, unbounded, satisfies $J_{k+1}(0) = 0$, and satisfies

$$J_{k+1}(r) \leq J_k(r) < r \quad (r > 0).$$

This proves the first six claims by induction on k .

Fix an integer threshold $t \geq 0$. Since J_k is unbounded and nondecreasing, the set $\{r \geq 0 : J_k(r) \leq t\}$ is finite. Since $J_k(0) = 0 \leq t$, it is nonempty. Hence $R_k(t)$ is a finite maximum. If $0 \leq t \leq t'$ are integers, then $\{r : J_k(r) \leq t\} \subseteq \{r : J_k(r) \leq t'\}$, so $R_k(t) \leq R_k(t')$. Since $J_{k+1}(r) \leq J_k(r)$, we also have $R_{k+1}(t) \geq R_k(t)$. $\square$

4.3 Threshold recurrences

Lemma 4.5 (Exact base inverse). *For every integer $t \geq 0$,*

$$R_0(t) = 2t + 1.$$

Proof. For integer $t \geq 0$, the inequality $J_0(r) \leq t$ is equivalent to

$$\left\lceil \frac{r-1}{2} \right\rceil \leq t,$$

which is equivalent to $r \leq 2t + 1$. Hence the largest such r is $2t + 1$. $\square$

Lemma 4.6 (Diamond-to-threshold recurrence). *Let B, t be nonnegative integers. If*

$$r \leq R_k(2^B)$$

and

$$B \leq R_{k+1}(t),$$

then

$$J_{k+1}(r) \leq t + 1.$$

Consequently,

$$R_{k+1}(t + 1) \geq R_k(2^{R_{k+1}(t)}).$$

Proof. Assume $r \leq R_k(2^B)$. Then $J_k(r) \leq 2^B$. If $J_k(r) \leq 1$, then $J_{k+1}(r) = J_k(r) \leq 1 \leq t + 1$. If $J_k(r) > 1$, then

$$J_{k+1}(r) = 1 + J_{k+1}(\lceil \log_2 J_k(r) \rceil).$$

Since $J_k(r) \leq 2^B$, we have $\lceil \log_2 J_k(r) \rceil \leq B$. By Lemma 4.4, J_{k+1} is nondecreasing. Since $B \leq R_{k+1}(t)$,

$$J_{k+1}(\lceil \log_2 J_k(r) \rceil) \leq J_{k+1}(B) \leq t.$$

Thus $J_{k+1}(r) \leq t + 1$.

Therefore every $r \leq R_k(2^B)$ satisfies $J_{k+1}(r) \leq t + 1$, so $R_{k+1}(t + 1) \geq R_k(2^B)$. Taking $B = R_{k+1}(t)$ proves the displayed recurrence. $\square$

Lemma 4.7 (Threshold jump). *For every $k \geq 0$ and every $Q \geq 2$,*

$$R_{k+1}(Q) \geq R_k(4Q).$$

Proof. Apply Lemma 4.6 with $t = Q - 1$:

$$R_{k+1}(Q) \geq R_k(2^{R_{k+1}(Q-1)}).$$

By Lemmas 4.4 and 4.5,

$$R_{k+1}(Q - 1) \geq R_0(Q - 1) = 2Q - 1.$$

For $Q \geq 2$,

$$2^{2Q-1} \geq 4Q.$$

Using monotonicity of R_k in its threshold argument,

$$R_{k+1}(Q) \geq R_k(2^{R_{k+1}(Q-1)}) \geq R_k(4Q).$$

□

4.4 Ackermann facts

Lemma 4.8 (Ackermann monotonicity and row domination). *For the Ackermann function A defined above:*

- (i) *for every $i \geq 0$, $A(i, x)$ is nondecreasing in x ;*
- (ii) *for every $i \geq 1$ and $x \geq 1$, $A(i, x) \geq 2x$;*
- (iii) *for every $i \geq 0$ and $x \geq 1$, $A(i + 1, x) \geq A(i, x)$.*

Proof. We first prove monotonicity in the second argument and the growth bound together by induction on the row index. For $i = 0$, monotonicity follows from $A(0, x) = 2x$.

Fix $i \geq 1$ and assume the corresponding lower-row facts have already been proved. We prove row- i monotonicity by induction on x . Since $A(i, 0) = 0$ and $A(i, 1) = 2$, the first step is clear. For $x \geq 1$,

$$A(i, x + 1) = A(i - 1, A(i, x)).$$

If $i = 1$, then $A(0, y) = 2y \geq y$ for $y \geq 0$. If $i \geq 2$, the lower-row growth bound gives $A(i - 1, y) \geq 2y \geq y$ for $y \geq 1$. Hence $A(i, x + 1) \geq A(i, x)$, and row i is nondecreasing.

We next prove the growth bound for row i . At $x = 1$, $A(i, 1) = 2 = 2x$. For $x \geq 2$, again using

$$A(i, x) = A(i - 1, A(i, x - 1)),$$

we get $A(i, x) = 2A(i, x - 1)$ when $i = 1$, and $A(i, x) \geq 2A(i, x - 1)$ when $i \geq 2$ by the lower-row growth bound. Thus, by induction on x with base $A(i, 1) = 2$,

$$A(i, x) \geq 2A(i, x - 1) \geq 4(x - 1) \geq 2x.$$

This proves monotonicity and growth for all rows.

For row domination, fix $i \geq 0$. At $x = 1$, $A(i + 1, 1) = 2 = A(i, 1)$. For $x \geq 2$,

$$A(i + 1, x) = A(i, A(i + 1, x - 1)).$$

The growth bound gives $A(i + 1, x - 1) \geq 2(x - 1) \geq x$, and row i is nondecreasing. Hence $A(i + 1, x) \geq A(i, x)$. □

Lemma 4.9 (Ackermann domination invariant). *For every $j \geq 1$ and $x \geq 1$,*

$$R_j(x) \geq A(j+1, x).$$

Proof. We prove the claim by induction on j , with an inner induction on x .

First let $j = 1$. For $x = 1$, Lemma 4.6 with $k = 0, t = 0$ gives

$$R_1(1) \geq R_0(2^{R_1(0)}).$$

By Lemma 4.4, $R_1(0) \geq R_0(0) = 1$. Thus

$$R_1(1) \geq R_0(2) = 5 \geq 2 = A(2, 1).$$

Assume $R_1(x) \geq A(2, x)$. By Lemma 4.6,

$$R_1(x+1) \geq R_0(2^{R_1(x)}).$$

Since $R_0(t) = 2t + 1$,

$$R_0(2^{R_1(x)}) = 2 \cdot 2^{R_1(x)} + 1.$$

Using the induction hypothesis and $A(1, y) = 2^y$,

$$2 \cdot 2^{R_1(x)} + 1 \geq 2^{A(2, x)} = A(2, x+1).$$

Thus the case $j = 1$ holds.

Now assume $R_j(x) \geq A(j+1, x)$ for all $x \geq 1$. We prove $R_{j+1}(x) \geq A(j+2, x)$ for all $x \geq 1$. For $x = 1$, Lemma 4.4 gives

$$R_{j+1}(1) \geq R_0(1) = 3 \geq 2 = A(j+2, 1).$$

Assume $R_{j+1}(x) \geq A(j+2, x)$. By Lemma 4.6,

$$R_{j+1}(x+1) \geq R_j(2^{R_{j+1}(x)}).$$

By the outer induction hypothesis,

$$R_j(y) \geq A(j+1, y) \quad (y \geq 1).$$

Thus

$$R_j(2^{R_{j+1}(x)}) \geq A(j+1, 2^{R_{j+1}(x)}).$$

By the inner inductive hypothesis,

$$R_{j+1}(x) \geq A(j+2, x).$$

Since $2^k > k$ for every integer $k \geq 0$, we have

$$2^{R_{j+1}(x)} > R_{j+1}(x) \geq A(j+2, x).$$

By Lemma 4.8(i), $A(j+1, -)$ is nondecreasing, so

$$A(j+1, 2^{R_{j+1}(x)}) \geq A(j+1, A(j+2, x)) = A(j+2, x+1),$$

where the last equality is the Ackermann recursion. Hence $R_{j+1}(x+1) \geq A(j+2, x+1)$. $\square$

4.5 Main comparison theorem

Theorem 4.10 (Main comparison). *For all integers $z \geq 1$ and $Q \geq 1$,*

$$R_{z+1}(Q) \geq A(z, 4Q).$$

Consequently, for every integer $r \geq 0$,

$$A(z, 4Q) > r \implies J_{z+1}(r) \leq Q.$$

Proof. First suppose $Q \geq 2$. By Lemma 4.7,

$$R_{z+1}(Q) \geq R_z(4Q).$$

Since $z \geq 1$, Lemma 4.9 gives

$$R_z(4Q) \geq A(z+1, 4Q).$$

By row domination from Lemma 4.8,

$$A(z+1, 4Q) \geq A(z, 4Q).$$

Therefore $R_{z+1}(Q) \geq A(z, 4Q)$.

Now suppose $Q = 1$ and $z = 1$. Lemma 4.6 gives

$$R_2(1) \geq R_1(2^{R_2(0)}).$$

By Lemma 4.4, $R_2(0) \geq R_0(0) = 1$, so $R_2(1) \geq R_1(2)$. The proof of Lemma 4.9 gave $R_1(1) \geq 5$, and Lemma 4.6 gives

$$R_1(2) \geq R_0(2^{R_1(1)}) \geq R_0(2^5) = 65.$$

Thus

$$R_2(1) \geq 65 \geq 16 = A(1, 4).$$

Finally suppose $Q = 1$ and $z \geq 2$. Lemma 4.6 gives

$$R_{z+1}(1) \geq R_z(2^{R_{z+1}(0)}).$$

By Lemma 4.4, $R_{z+1}(0) \geq R_0(0) = 1$, so $R_{z+1}(1) \geq R_z(2)$. Apply Lemma 4.7 with level $z-1$ and $Q = 2$:

$$R_z(2) \geq R_{z-1}(8).$$

Because $z-1 \geq 1$, Lemma 4.9 gives

$$R_{z-1}(8) \geq A(z, 8).$$

By Ackermann monotonicity,

$$A(z, 8) \geq A(z, 4).$$

Therefore $R_{z+1}(1) \geq A(z, 4)$.

The three cases prove $R_{z+1}(Q) \geq A(z, 4Q)$ for all $z \geq 1$ and $Q \geq 1$. If $A(z, 4Q) > r$, then $r \leq A(z, 4Q) - 1 \leq R_{z+1}(Q)$, and therefore $J_{z+1}(r) \leq Q$. $\square$

5 Alpha consequences and cost bound

Definition 5.1 (Packet inverses). *Define*

$$\alpha_J^Q(m, n) = \min\{k \geq 0 : J_k(L(n)) \leq Q(m, n)\},$$

and

$$\alpha_Q(m, n) = \min\{z \geq 1 : A(z, 4Q(m, n)) > L(n)\}.$$

Lemma 5.2 (Existence of α_Q). *For all positive integers m, n , the minimum defining $\alpha_Q(m, n)$ exists.*

Proof. Since $Q(m, n) \geq 1$, we have $4Q(m, n) \geq 4$. Let $a_z = A(z, 4)$. Then $a_1 = 16$, and for $z \geq 1$,

$$A(z+1, 2) = 4, \quad A(z+1, 3) = A(z, 4) = a_z,$$

so

$$a_{z+1} = A(z+1, 4) = A(z, a_z).$$

By the growth bound in Lemma 4.8, $A(z, a_z) \geq 2a_z$. Hence a_z is unbounded in z . By monotonicity in the second argument,

$$A(z, 4Q(m, n)) \geq A(z, 4) = a_z,$$

so some $z \geq 1$ satisfies $A(z, 4Q(m, n)) > L(n)$. □

Theorem 5.3 (Patched threshold consequence). *For all positive integers m, n ,*

$$\alpha_J^Q(m, n) \leq \alpha_Q(m, n) + 1.$$

Proof. Let $z = \alpha_Q(m, n)$, $Q = Q(m, n)$, and $L = L(n)$. By definition,

$$A(z, 4Q) > L.$$

By Theorem 4.10,

$$R_{z+1}(Q) \geq A(z, 4Q) > L.$$

Since L is an integer, $L \leq R_{z+1}(Q)$. Therefore $J_{z+1}(L) \leq Q$, and hence

$$\alpha_J^Q(m, n) \leq z + 1 = \alpha_Q(m, n) + 1. \quad \square$$

Definition 5.4 (Source-faithful real-threshold inverse). *Define*

$$\alpha_J^S(m, n) = \min \left\{ k \geq 0 : J_k(L(n)) \leq 1 + \frac{m}{n} \right\}.$$

Lemma 5.5 (Ackermann buffer). *For $z \geq 1$ and integer $p \geq 1$,*

$$A(z+1, 4p) \geq A(z, 4p+4).$$

Proof. Since $4p \geq 2$,

$$A(z+1, 4p) = A(z, A(z+1, 4p-1)).$$

We claim $A(z+1, 4p-1) \geq 4p+4$. If $p=1$, then $4p-1=3$, and $A(i, 2)=4$ for all $i \geq 1$, so

$$A(z+1, 3) = A(z, 4) \geq 8 = 4p+4.$$

If $p \geq 2$, Lemma 4.8 gives

$$A(z+1, 4p-1) \geq 2(4p-1) = 8p-2 \geq 4p+4.$$

By monotonicity of $A(z, -)$, $A(z+1, 4p) \geq A(z, 4p+4)$. □

Theorem 5.6 (Source-faithful real-threshold consequence). *For all positive integers m, n ,*

$$\alpha_J^S(m, n) \leq \alpha_Q(m, n) + 2.$$

If $1 + m/n$ is integral, then

$$\alpha_J^S(m, n) \leq \alpha_Q(m, n) + 1.$$

Proof. Let

$$s = 1 + \frac{m}{n}, \quad Q = \lceil s \rceil, \quad L = L(n), \quad z = \alpha_Q(m, n).$$

Then $A(z, 4Q) > L$.

If s is an integer, then $Q = s$. Theorem 4.10 gives

$$R_{z+1}(s) = R_{z+1}(Q) \geq A(z, 4Q) > L.$$

Thus $J_{z+1}(L) \leq s$, and $\alpha_J^S(m, n) \leq z+1 = \alpha_Q(m, n) + 1$.

If s is not integral, put $p = \lfloor s \rfloor$. Since $m, n \geq 1$, $p \geq 1$, and $Q = p+1$. Thus $A(z, 4p+4) > L$. By Lemma 5.5, $A(z+1, 4p) > L$. Theorem 4.10 gives

$$R_{z+2}(p) \geq A(z+1, 4p) > L.$$

Thus $J_{z+2}(L) \leq p < s$, and therefore

$$\alpha_J^S(m, n) \leq z+2 = \alpha_Q(m, n) + 2.$$

□

Theorem 5.7 (Cost consequence). *Using the source recurrence $f(m, n, r) \leq km + 2nJ_k(r)$, one obtains*

$$f(m, n, L(n)) \leq (\alpha_Q(m, n) + 3)m + 4n.$$

Consequently,

$$f(m, n, L(n)) = O(m\alpha_Q(m, n) + n).$$

Proof. Let $z = \alpha_Q(m, n)$, $Q = Q(m, n)$, and $L = L(n)$. By Theorem 5.3, $J_{z+1}(L) \leq Q$. Applying the source recurrence with $k = z+1$ and $r = L$ gives

$$f(m, n, L) \leq (z+1)m + 2nJ_{z+1}(L) \leq (z+1)m + 2nQ.$$

Since

$$Q = \left\lceil 1 + \frac{m}{n} \right\rceil \leq 2 + \frac{m}{n},$$

we have

$$2nQ \leq 2n \left(2 + \frac{m}{n} \right) = 4n + 2m.$$

Therefore

$$f(m, n, L) \leq (z + 3)m + 4n.$$

Substituting $z = \alpha_Q(m, n)$ proves the claim. $\square$

6 Normalization discussion

Definition 6.1 (Classical row inverse for this normalization). *Define*

$$\alpha_A(N, X) = \min\{z \geq 1 : A(z, X) > N\}.$$

Then the packet inverse is exactly

$$\alpha_Q(m, n) = \alpha_A(L(n), 4Q(m, n)).$$

The proved comparison says

$$\alpha_J^Q(m, n) \leq \alpha_A(L(n), 4Q(m, n)) + 1.$$

For the source-faithful real threshold,

$$\alpha_J^S(m, n) \leq \alpha_A(L(n), 4Q(m, n)) + 2,$$

with the +1 improvement when $1 + m/n$ is integral.

The source defines

$$\alpha_S(m, n) = \min \left\{ k : J_k(\lceil \log_2 n \rceil) \leq 1 + \frac{m}{n} \right\}.$$

For $n \geq 2$, $L(n) = \lceil \log_2 n \rceil$, so α_J^S is the packet notation for this source threshold. For $n = 1$, the packet uses $L(1) = 1$ rather than 0 as a harmless stabilization.

The Seidel-Sharir paper discusses Tarjan's initial inverse-Ackermann function α_T , defined through a different hierarchy T_k [?, ?]. This paper does not use that comparison as a proof dependency. The theorem here is restricted to the concrete normalization $\alpha_Q(m, n) = \alpha_A(L(n), 4Q(m, n))$.

7 Hazards avoided

7.1 The threshold $1 + m/n$ versus $2m/n$

The source-faithful J threshold is

$$J_k(\lceil \log_2 n \rceil) \leq 1 + \frac{m}{n},$$

not a J_k threshold $2m/n$. The alternate condition

$$J_k(\lg n) \leq \frac{2m}{n}$$

is not source-faithful under the current packet unless independently verified.

Moreover, it is impossible for $m/n < 1/2$ and $n \geq 3$. Indeed, $n \geq 3$ gives $L(n) \geq 2$, and a direct induction on k gives $J_k(r) \geq 1$ for every $r \geq 2$: the claim is clear for J_0 , and the $g^\diamond$ definition either returns 1 in the small branch or $1 +$ something in the recursive branch. If $m/n < 1/2$, then $2m/n < 1$, so the alternate threshold would force $J_k(L(n)) = 0$, contradicting $J_k(L(n)) \geq 1$.

7.2 The factor $2n$ in the shifting lemma

The source shifting lemma has premise

$$f(m, n, r) \leq km + 2ng(r)$$

and conclusion

$$f(m, n, r) \leq (k + 1)m + 2ng^\diamond(r).$$

The earlier shorthand with $ng(r)$ is not used here.

7.3 Why $Q(m, n) = \lceil 1 + m/n \rceil$ is used

The comparison theorem is stated for an integer threshold $Q \geq 1$, while the source threshold $1 + m/n$ is real-valued. The packet threshold

$$Q(m, n) = \left\lceil 1 + \frac{m}{n} \right\rceil$$

provides an integer target for the R_k comparison. The exact real source threshold is recovered with one extra buffer in the nonintegral case.

7.4 Avoiding bottom-up Tarjan-potential drift

The proof uses only the Seidel–Sharir top-down J hierarchy, the $g^\diamond$ threshold recurrence, the packet Ackermann normalization, and the source recurrence $f(m, n, r) \leq km + 2nJ_k(r)$. It does not use bottom-up Tarjan potentials.

8 Conclusion

The direct bridge is

$$R_{z+1}(Q) \geq A(z, 4Q),$$

with constants $c = 1$, $C = 1$, and $D = 4$ in the comparison target. This bridge converts the Seidel–Sharir J hierarchy into the packet-normalized inverse-Ackermann bound. The result is proof-digestion and exposition: it does not claim a new union-find upper bound, but it makes the Ackermann comparison explicit and source-faithful. It is presented as a source-faithful answer to Tarjan’s Dagstuhl 25191 §5.3 comparison prompt rather than a priority claim.

Corollary 8.1 (Final citable theorem block). *Assume the Seidel–Sharir J hierarchy and the packet Ackermann normalization above. Then, for all integers $z \geq 1$ and $Q \geq 1$,*

$$R_{z+1}(Q) \geq A(z, 4Q).$$

Thus the comparison target $R_{cz+C}(Q) \geq A(z, DQ)$ holds with

$$c = 1, \quad C = 1, \quad D = 4.$$

Let

$$L(n) = \lceil \log_2 \max(n, 2) \rceil, \quad Q(m, n) = \left\lceil 1 + \frac{m}{n} \right\rceil,$$

and define α_Q , α_J^Q , and α_J^S as above. Then

$$\alpha_J^Q(m, n) \leq \alpha_Q(m, n) + 1,$$

$$\alpha_J^S(m, n) \leq \alpha_Q(m, n) + 2,$$

with the sharper bound $\alpha_J^S(m, n) \leq \alpha_Q(m, n) + 1$ when $1 + m/n$ is integral. Finally,

$$f(m, n, L(n)) \leq (\alpha_Q(m, n) + 3)m + 4n,$$

and hence

$$f(m, n, L(n)) = O(m\alpha_Q(m, n) + n).$$

A Source-anchor checklist

This appendix records the source anchors used in the proof. Page references are to the author PDF / published pagination used in the source-fidelity packet; they are included to make the proof-exposition dependencies auditable.

- **Seidel–Sharir, page 7.** This page contains the rank bound used to justify the logarithmic rank cutoff, the definitions of g^* and $g^\diamond$, and the statement of Lemma 5, the top-down shifting lemma.
- **Seidel–Sharir, page 8.** This page completes the proof of Lemma 5, which is the source-correct shifting lemma with the factor $2n$.
- **Seidel–Sharir, page 9.** This page contains Corollary 6, including the J hierarchy, the recurrence $f(m, n, r) \leq km + 2nJ_k(r)$, and the source inverse threshold α_S defined through $J_k(\lceil \log_2 n \rceil) \leq 1 + m/n$.
- **Dagstuhl Seminar 25191, Section 5.3.** This section records the public comparison prompt motivating the direct relationship between the Seidel–Sharir top-down hierarchy and Ackermann-style inverse bounds.

The paper uses these anchors only for source setup. The comparison theorem $R_{z+1}(Q) \geq A(z, 4Q)$ is proved internally from the J hierarchy and the packet Ackermann normalization.

References

- [1] Raimund Seidel and Micha Sharir. Top-down analysis of path compression. *SIAM Journal on Computing*, 34(3):515–525, 2005. DOI: <https://doi.org/10.1137/S0097539703439088>; author PDF: <https://www.math.tau.ac.il/~michas/ufind.pdf>.
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